ECON 0100 | HW E2 | Competitive Firms

Treat Homework as a tool to highlight what you don't yet understand. Grading is by submission, so there's no additional points for turning in perfect answers you don't fully understand. Use your areas of confusion to guide your study. Work in a group and ask others for help.

Q1 | A Price for Brooms

In the early days after the discovery of autolocomotion, SwiftTech's method for broomstick manufacturing spread widely. Many identical makers began selling the same standard broom, each one taking the market price. Each maker's fixed cost is 360 coins a week, with a marginal cost of:

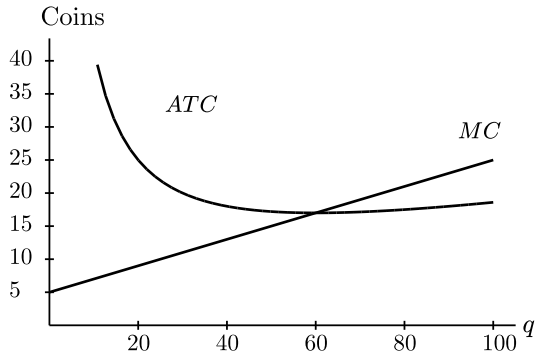
$$MC = 5 + \frac{q}{5}$$

where q is the number of brooms it makes in a week. The table shows its average total cost at some quantities. Each graph shows one maker's marginal cost and average total cost.

Brooms	**20**	**40**	**60**	**90**
ATC	25	18	17	18

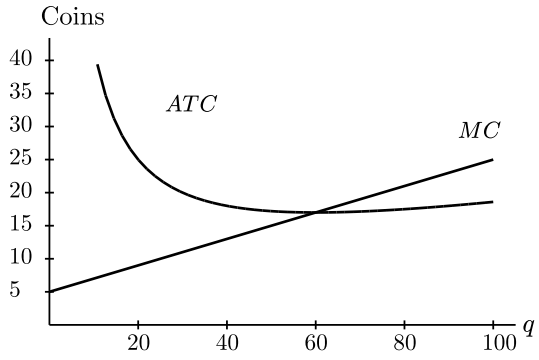
If the market price is 23 coins:

a) Find and graph marginal revenue (MR).
b) Find and graph the firm's best quantity (q^*).
c) Find and graph the firm's maximized profit.



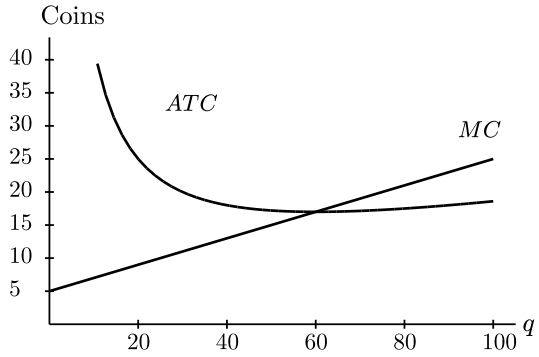
If the price falls to 13 coins:

d) Find and graph the firm's q^* if they stay open.
e) Find and graph the firm's maximized profit if open.
f) What is the firm's profit / loss if they shutdown?
g) Should the firm stay open or shutdown?



In the long run:

h) Do firms enter or exit at a price of 23 coins?
i) Do firms enter or exit at a price of 13 coins?
j) Find and graph the long-run price.
k) Find and graph the firm's q^* and profit at that price.



Q2 | Subsidizing Broomsticks

Quidditch is a game much beloved by the magical community. Since the rules are well established, the game equipment is standardized, and every equipment seller produces identical equipment. Broomsticks are an important part of the game, and anyone who wants to can start making and selling them. The number of sellers of broomsticks hasn't changed much in the over 500 years since SwiftTech's methods spread.

To spur interest in the sport, the Ministry imposed a subsidy on broomsticks. Using a couple of graphs to illustrate your answer, explain what happened in the market because of this subsidy. Be sure to include all relevant components on the graph, and conclude in a long-run equilibrium.

The Market



One Firm

